

- 5 a Energy is dissipated in the internal resistance of the battery. **B1**
- b $E = V + Ir$ **B1**
- ci The graph shows that when current from the battery is zero, V is equal to E (which is 2.8 V). **B1**
- cii Either Or
- When $V = 1.4 \text{ V}$, $I = 1.0 \text{ A}$ $r = - \text{gradient}$
- $E = V + Ir$ $= - \frac{2.8 - 1.4}{0 - 1.0}$ **C1**
- $2.8 = 1.4 + 1.0r$ $r = 1.4 \Omega$ **A1**
- di When $V = 2.1 \text{ V}$, $I = 0.50 \text{ A}$ **B1**
- $V = IR$
- $2.1 = 0.50 R$
- $R = 4.2 \Omega$ **A1**
- dii $I = \frac{Q}{t} = \frac{N \times e}{t}$
- $0.50 = \frac{N \times 1.6 \times 10^{-19}}{1.0}$ **C1**
- $N = 3.1 \times 10^{18} \text{ electrons}$ **A1**
- diii Energy = EIt
- $(9.2 - 1.6) \times 10^3 = 2.8 \times 0.50 \times t$ **C1**
- $t = 5.4 \times 10^3 \text{ s}$ **A1**